# Practice Questions Sheet 1

This tutorial introduces some **basic ways of describing and visualising data** in R. Please **attempt all questions** before your tutorial You should work through the questions in a new R script. For questions asking you to interpret a result, **aim for one or two clear sentences**. You should use the context of the pizza delivery data when explaining your answers.

Student name: _____

Student number: _____

**Please complete this assignment before the tutorial on the week of 12th October**

*If you cannot remember what a certain function does or a certain function's argument*

If you are unsure what a function does, remember that you can use R's help system:

```
# What does the mean function do?
?mean()
```

You can also use the Help pane in RStudio.

# Pizza delivery data

We will use a dataset containing information about **1,301 pizza deliveries**.
The dataset contains **four variables**:

Table 1: Overview of the pizza dataset

| Variable | Description |
|---|---|
| Day | Day of the week on which the delivery was made |
| Distance | Distance between the pizza restaurant and delivery location, in km |
| Order amount | Value of the pizza order |
| Tip | Tip left by the customer |

The dataset is available on the RStudio server in the ST201 data folder.
You can load it by opening a new R script and running the following code:

```
# Load the pizza dataset
pizza <- readRDS("Sharedfiles/ST201/data/pizza_del.rds")
```

Once you have loaded the data, you can look at the first few observations using:

```
# Examine the first 6 rows of the pizza dataset
head(pizza)
```

## Exercise 1

- How many **observations** are in the dataset?
- How many **variables** are in the dataset?
- Which variables are **categorical** and which are **numerical**?

# Individual observations

Let's start by looking at the `distance` variable.
Copy and run the below code

```
pizza$distance[1:10]
```

This displays the first 10 observations of distance.

> Remember that R uses square brackets `[]` to select observations.
>
> For example
>
> ```
> # The first observation
> pizza$distance[1]
> ```
>
> selects the **first observation**, while
>
> ```
> # The tenth observation
> pizza$distance[10]
> ```
>
> selects the **tenth observation**.

### Exercise 2

- What is the value of the **275th observation** of `distance`?
- What does this **value represent** in the context of the dataset?
- What is the distance for the **1,000th observation**?

# Measuring the centre of the data

Suppose the owner of the pizza restaurant wants to know the typical delivery distance.
One way to describe the centre of a numerical variable is using the mean.
In Figure 1, we can see the full distribution of the pizza delivery distance variable.




Figure 1: Histogram showing the distribution of pizza delivery distance (km)

To start off, copy and run the below code

```
mean(pizza$distance)
```

### Exercise 3

- What is the **mean** delivery distance?
- Modify the above code and calculate the **median** delivery distance?
- Write one or two short sentences comparing the mean and median and then in the context of the data, compare the two values (are they similar, or noticeably different)

# Quantiles

The mean and median tell us something about the centre of the data.
We can also use **quantiles** to describe where observations fall within the distribution.
Run the below code

```
quantile(pizza$ditance, )
```

**Aha!! You should get an warning**

### Exercise 4

- Modify the above code to correctly calculate the 25th percentile
- Following this, calculate the 75th percentile
- Provide a short interpretation of these two quantiles, in the context of the pizza deliveries

# Measuring variance

So far, we've been exploring the centre of `distance`. Let's zoom out a little and look at the spread of our variable

### Exercise 5

- Calculate the **variance** of the `distance` variable
- Now, calculate the **standard deviation** of the `distance` variable
- In general terms, what is the difference between the term *variance* and the term *standard deviation*
- What is the **range** of the `distance` variable

# Visualising data

Figure 2 is a box plot of the pizza `tips` variable

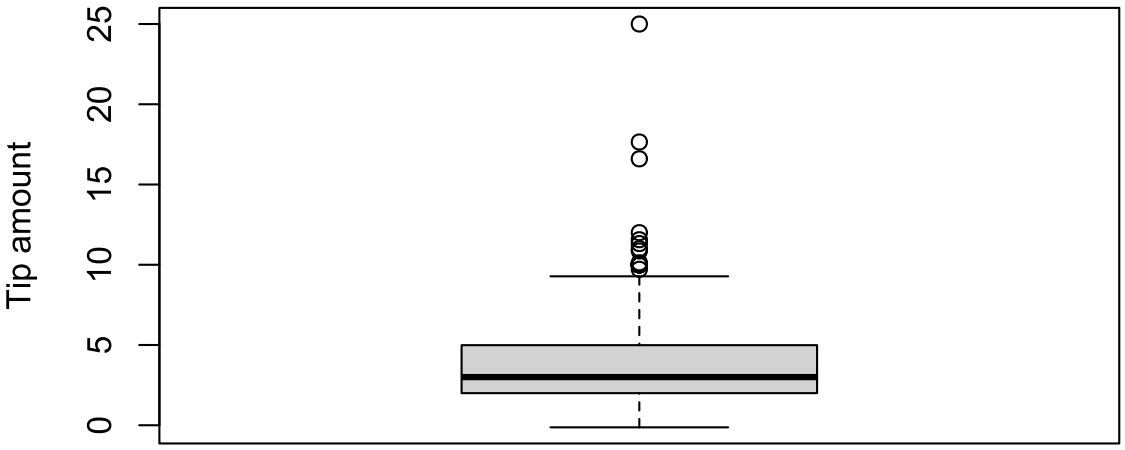



Figure 2: Boxplot of pizza tips

### Exercise 6

- Write `R` code to produce the above plot (remember to add in the **y label**)
- Provide a short overview of what the boxplot is telling us (what is the median of the `tip` variable, what is the range, are there any outliers, et.)
- Modify the `R` code you just wrote to produce a histogram of the `tip` variable

# Comparing delivery days

So far, we have been examining the data across all deliveries.
However, we can actually split this data up a little and look at each separate day (with the use of the `day` variable).
Suppose the restaurant owner wants to know whether customers spend different amounts on different days of the week.
Run the below code

```
# Create a dataset only with pizza deliveries on a Saturday
sat_pizza <- pizza[pizza$day == "Saturday", ]

head(sat_del)
```

This creates a new dataset containing Saturday deliveries only.

### Exercise 7

- Using the new dataset, calculate the mean `order_amount` for Saturday deliveries.
- Create a dataset containing **Monday** deliveries only.
- Calculate the mean `order_amount` for Monday deliveries.
- Which day has the larger mean order amount?

**i** Bonus question

Creating a bunch of new datasets with different delivery days can be quite inefficient and tedious.
*Imagine doing it 7 times and running seven different analyses!*
Instead, a neat thing we can do is compare all the deliveries at once
Copy and paste the below code which outputs Figure 3

```
boxplot(tip ~ day, data = pizza, ylab = "Tip amount")
```

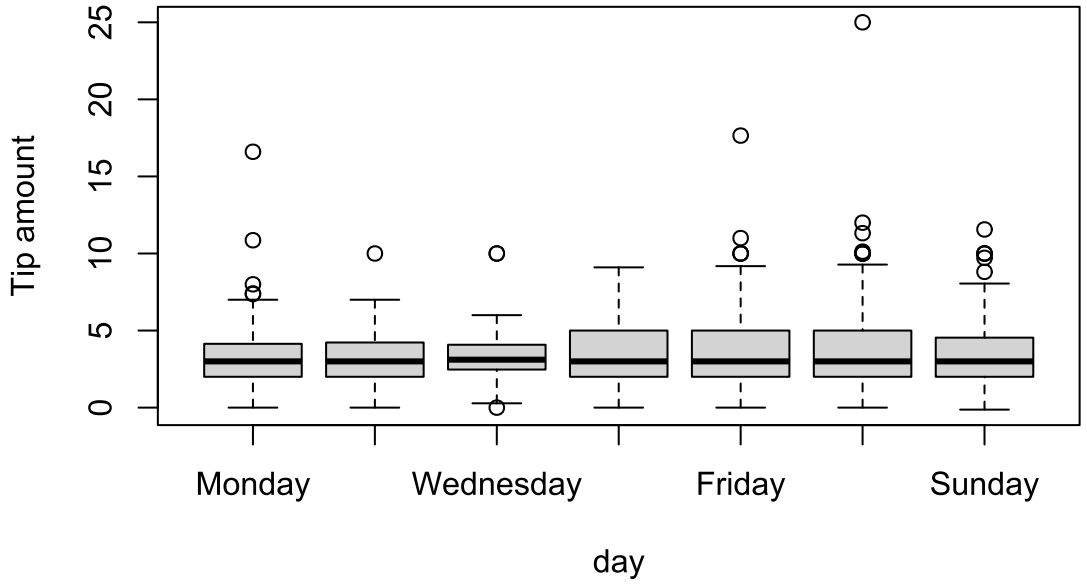



Figure 3: Overview of pizza tips across all delivery days

The ~ tells R to "*Plot tips separately for each value of day*".

- Which day appears to have the **lowest variation in tips**?
- Are there any potential outliers?